Fundamentals of Electrical and Electronics Engineering (FEEE)

SUB CODE: ECE1002

T-P-C: 3-2-4

Lecture-25

Faculty

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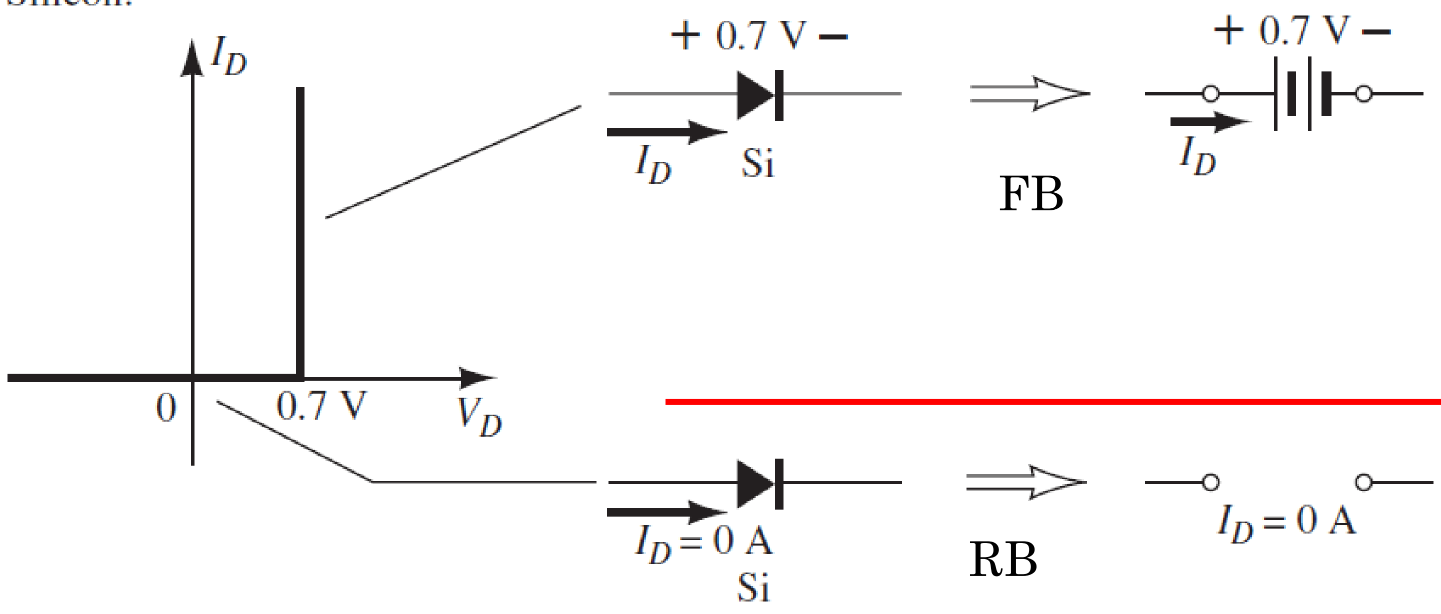
Diode Equivalent:

ON State(If current exists in loop by source) then replace with Vd = Forward bias

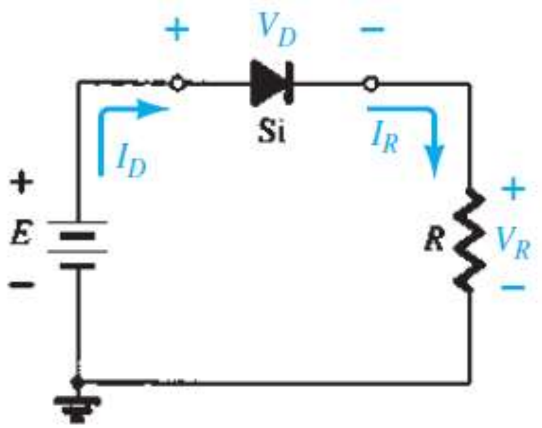


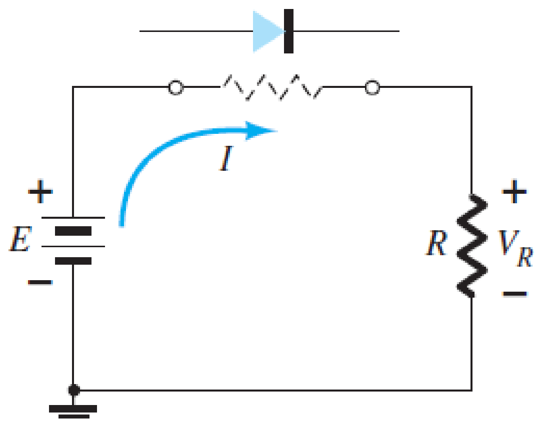
Special Case:(No bias)
If current(I) NOT exists by source in loop then replace Vd=0(short circuit)→No bias condition.

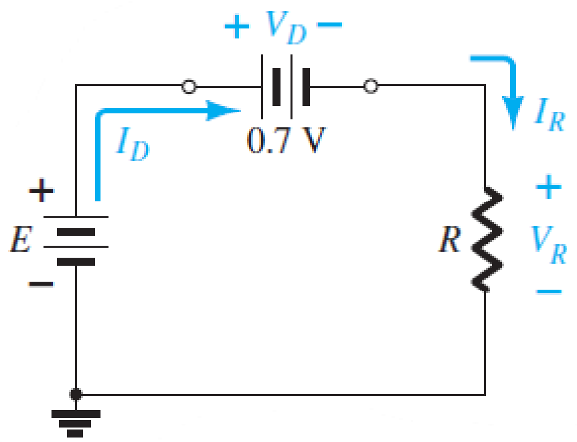
OFF State
(Open Circuit)=Reverse bias

- **In general, a diode is in the “on” state if the current established by the applied sources is such that its direction matches that of the arrow in the diode symbol, and V_D > 0.7 V for silicon, V_D >0.3 V for germanium, and V_D > 1.2 V for gallium arsenide.**
- If no current exists in loop then I =0 and V=0, leads to diode short circuit.

CASE-1





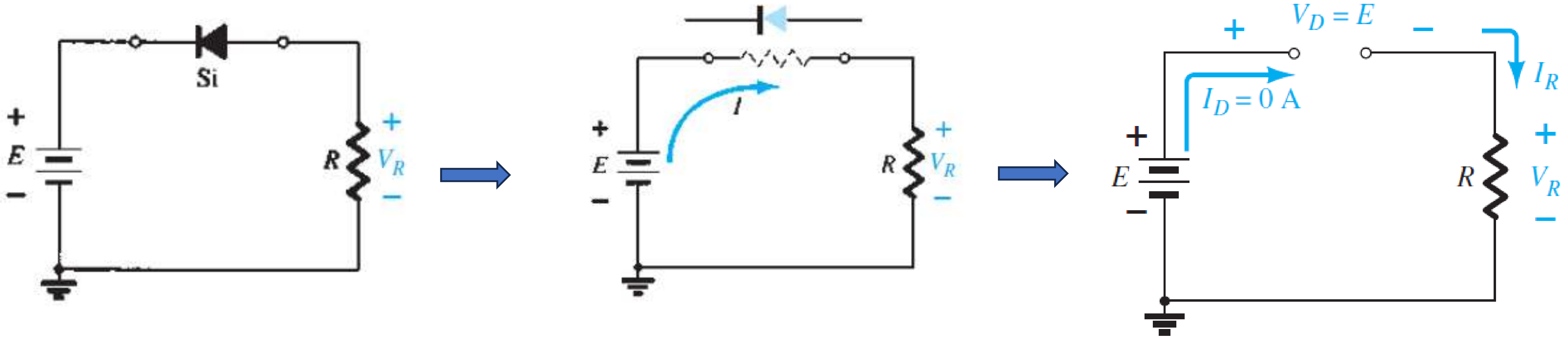


$$\boxed{V_D = V_K}$$

$$\boxed{V_R = E - V_K}$$

$$\boxed{I_D = I_R = \frac{V_R}{R}}$$

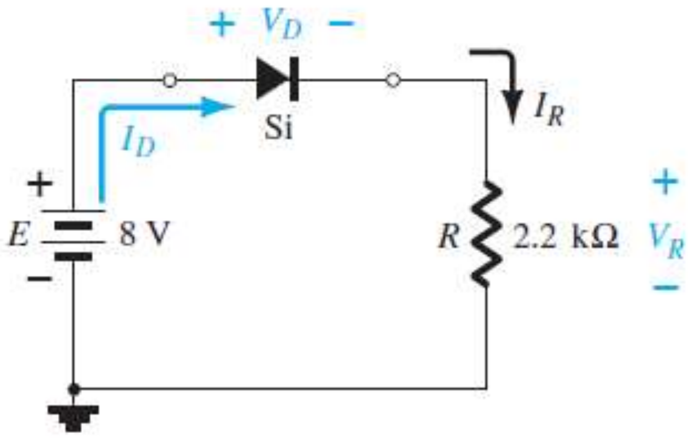
CASE-2



$$V_R = I_R R = I_D R = (0\text{ A})R = \mathbf{0\ V}$$

II).Problems on voltage across diode(Si /Ge) and Current through diode with DC source

1) For the series diode configuration of Fig below, determine V_D , V_R , and I_D .



- Since the applied voltage establishes a current in the clockwise direction to match the arrow of the symbol and the diode is in the "on" state,

$$V_D = \mathbf{0.7\ V}$$

$$V_R = E - V_D$$

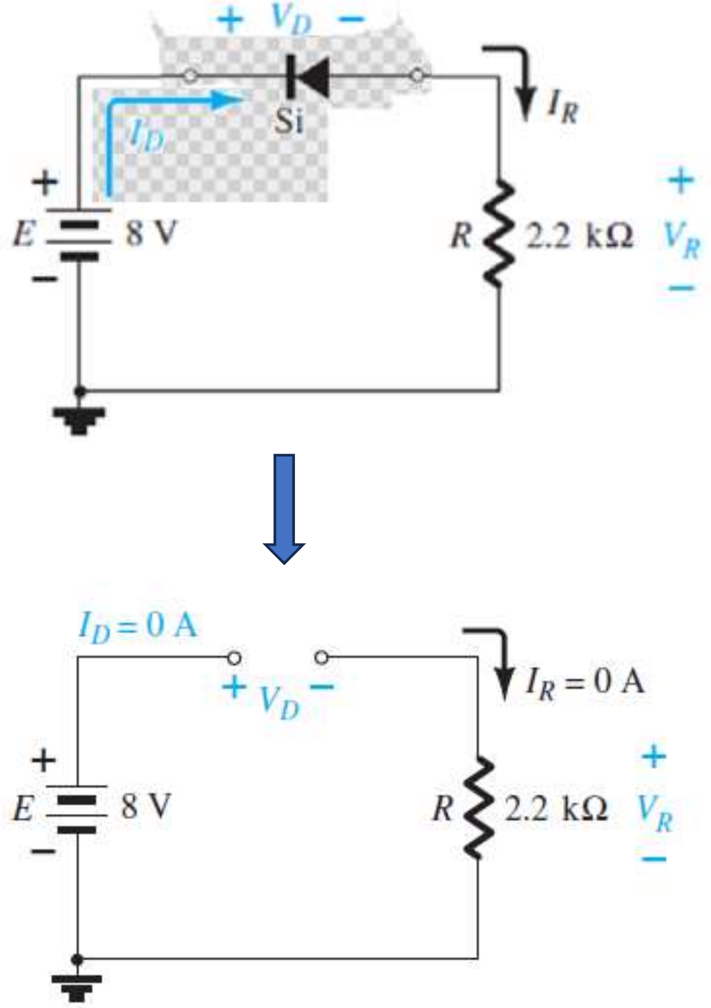
$$= 8\text{ V} - 0.7\text{ V} = \mathbf{7.3\ V}$$

$$I_D = I_R = \frac{V_R}{R} = \frac{7.3\text{ V}}{2.2\text{ k}\Omega} \cong \mathbf{3.32\ mA}$$

2) For the series diode configuration of Fig below, determine V_D , V_R , and I_D .

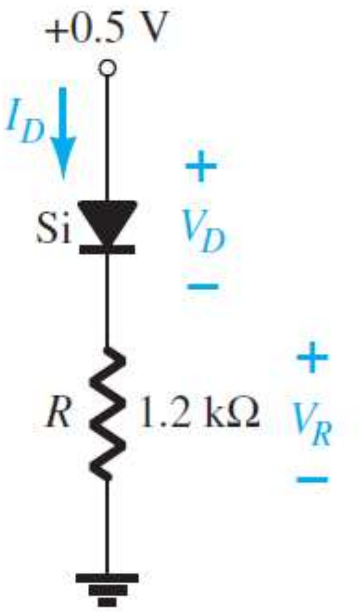
$$E - V_D - V_R = 0$$

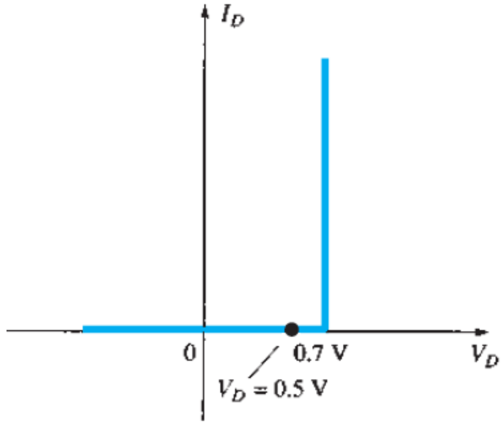
$$V_D = E - V_R = E - 0 = E = 8\text{ V}$$

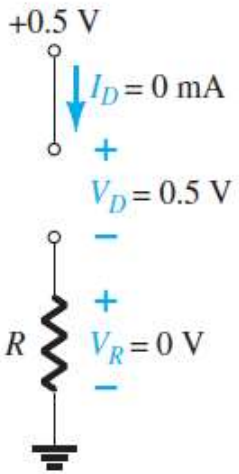


3) For the series diode configuration of Fig below, determine V_D , V_R , and I_D .

- Although the "pressure" establishes a current with the same direction as the arrow symbol, the level of applied voltage is insufficient to turn the silicon diode "on."

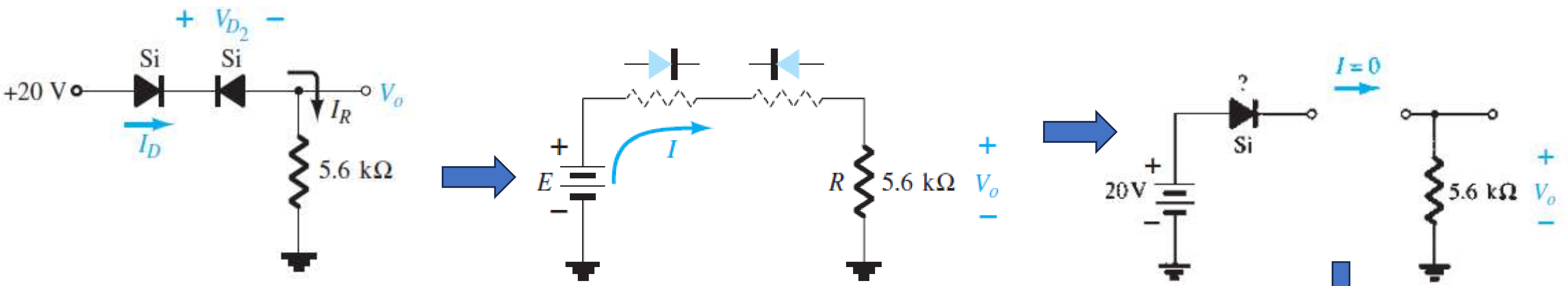






$$I_D = \mathbf{0\ A}$$
$$V_R = I_R R = I_D R = (0\ \text{A})\, 1.2\ \text{k}\Omega = \mathbf{0\ V}$$
$$V_D = E = \mathbf{0.5\ V}$$

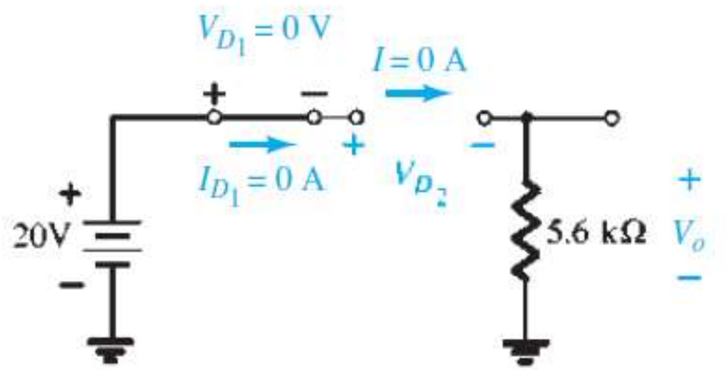
4)Determine I_D , V_{D2}, and V_o for the circuit of Fig. below.



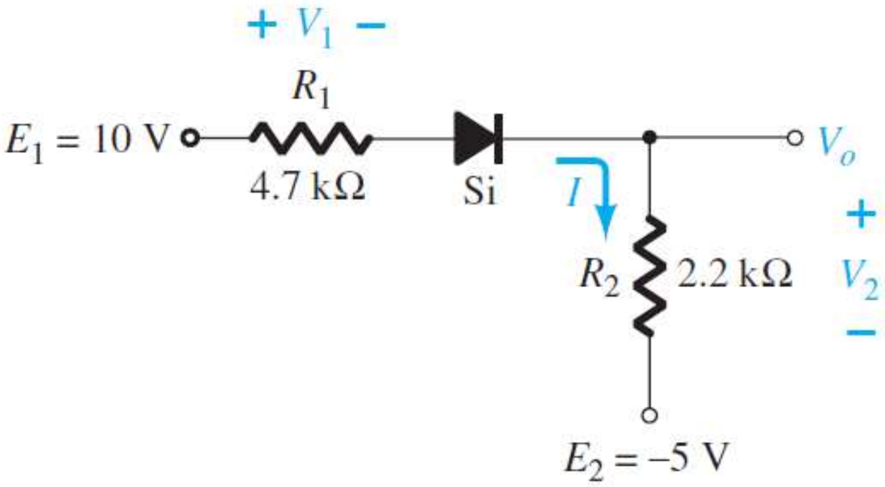
$$I_D = 0 \text{ A and } V_{D_1} = 0 \text{ V}$$

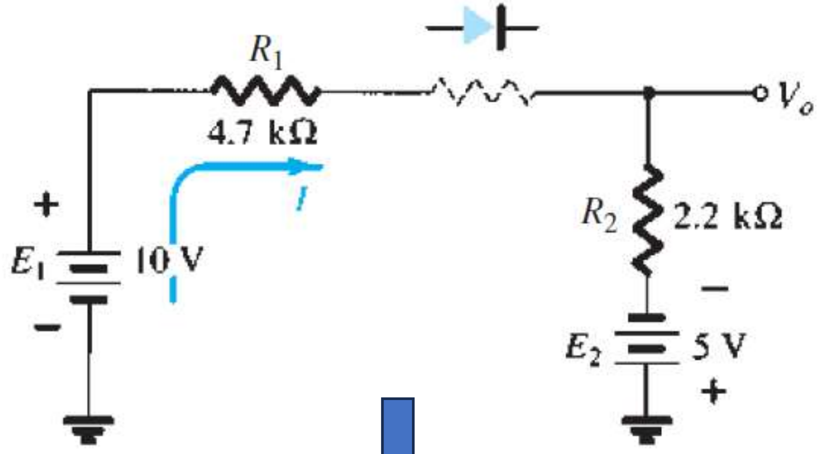
$$V_{D_2} = V_{\text{open circuit}} = E = \mathbf{20\ V}$$

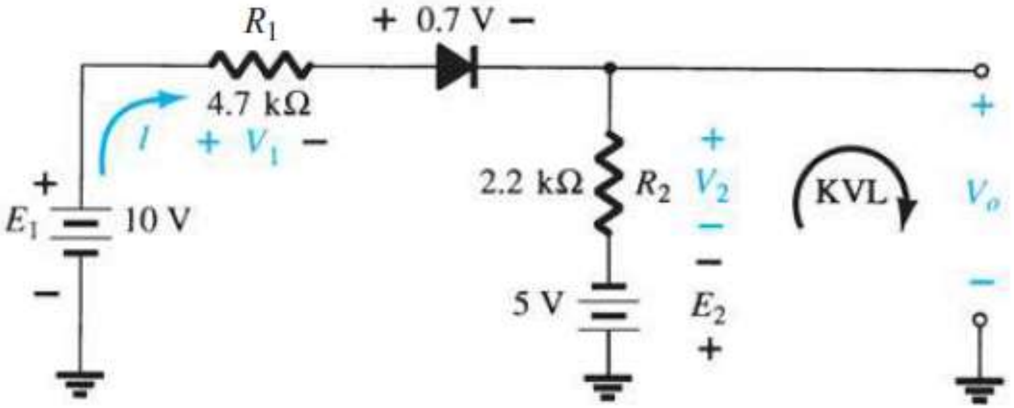
$$V_o = I_R R = I_D R = (0\text{ A})R = \mathbf{0\ V}$$



5) Determine I , V_1 , V_2 , and V_o for the series dc configuration of Fig below.







$$I = \frac{E_1 + E_2 - V_D}{R_1 + R_2} = \frac{10\text{ V} + 5\text{ V} - 0.7\text{ V}}{4.7\text{ k}\Omega + 2.2\text{ k}\Omega} = \frac{14.3\text{ V}}{6.9\text{ k}\Omega}$$

$$\cong \mathbf{2.07\ mA}$$

$$V_1 = IR_1 = (2.07\text{ mA})(4.7\text{ k}\Omega) = \mathbf{9.73\ V}$$

$$V_2 = IR_2 = (2.07\text{ mA})(2.2\text{ k}\Omega) = \mathbf{4.55\ V}$$

$$-E_2 + V_2 - V_o = 0$$

$$V_o = V_2 - E_2 = 4.55\text{ V} - 5\text{ V} = \mathbf{-0.45\ V}$$